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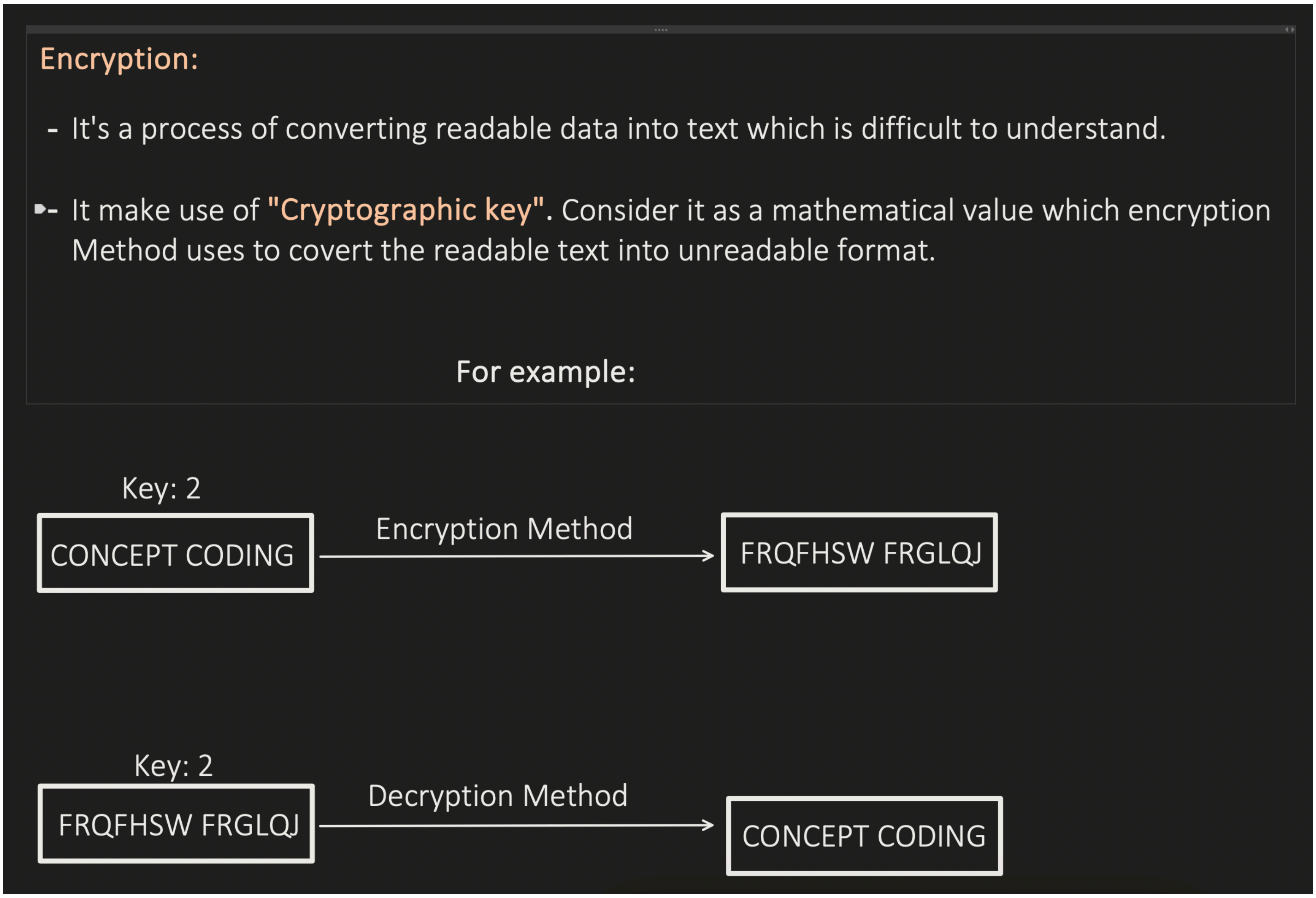
# Cryptography

**"Concept && Coding" YT Video Notes**

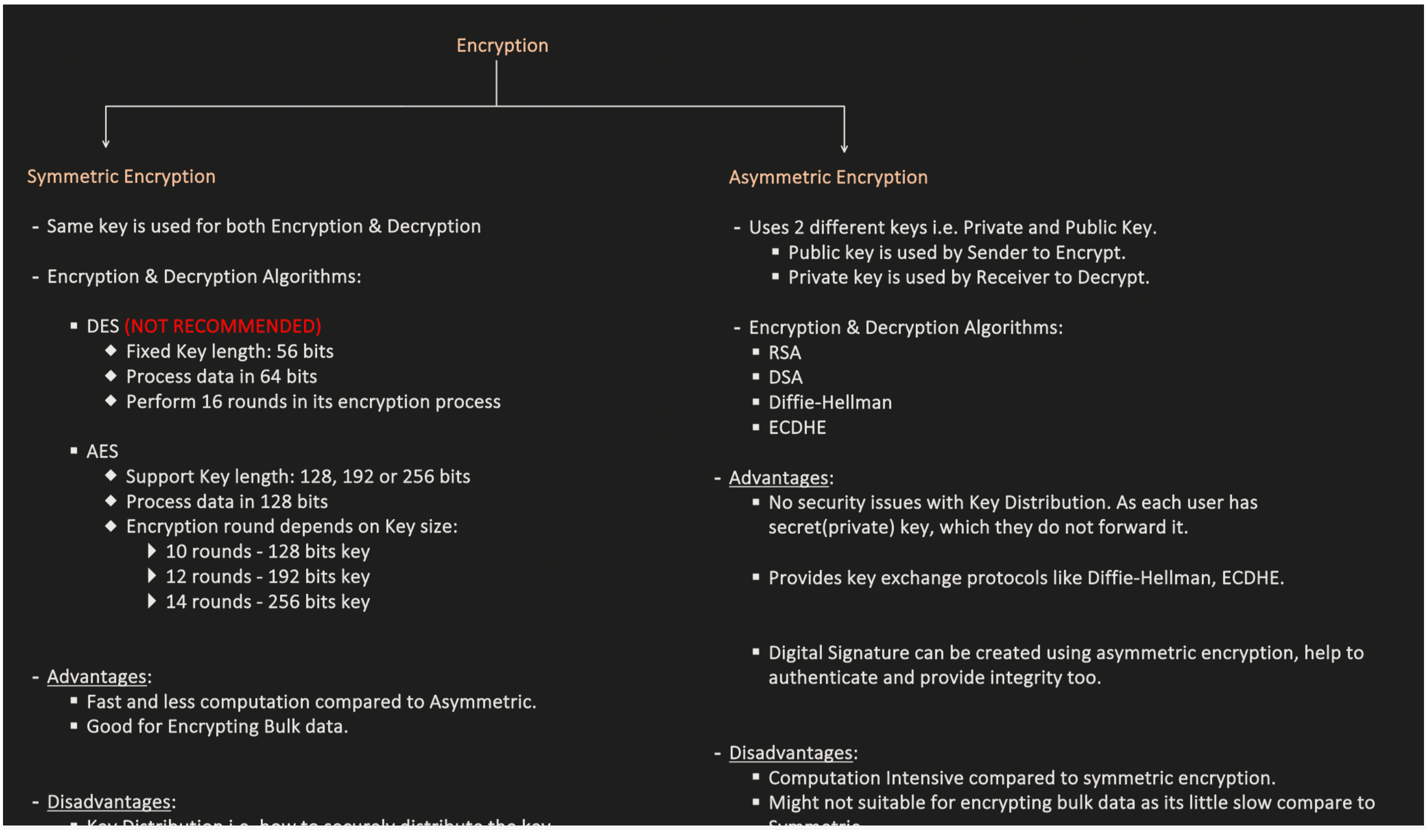


**Encryption:**

- It's a process of converting readable data into text which is difficult to understand.
- It make use of **"Cryptographic key"**. Consider it as a mathematical value which encryption Method uses to covert the readable text into unreadable format.

For example:





Encryption

**Symmetric Encryption**

- Same key is used for both Encryption & Decryption
- Encryption & Decryption Algorithms:
  - DES (NOT RECOMMENDED)
    - Fixed Key length: 56 bits
    - Process data in 64 bits
    - Perform 16 rounds in its encryption process
  - AES
    - Support Key length: 128, 192 or 256 bits
    - Process data in 128 bits
    - Encryption round depends on Key size:
      - 10 rounds - 128 bits key
      - 12 rounds - 192 bits key
      - 14 rounds - 256 bits key
- Advantages:
  - Fast and less computation compared to Asymmetric.
  - Good for Encrypting Bulk data.
- Disadvantages:
  - Key Distribution i.e. how to securely distribute the key.

**Asymmetric Encryption**

- Uses 2 different keys i.e. Private and Public Key.
  - Public key is used by Sender to Encrypt.
  - Private key is used by Receiver to Decrypt.
- Encryption & Decryption Algorithms:
  - RSA
  - DSA
  - Diffie-Hellman
  - ECDHE
- Advantages:
  - No security issues with Key Distribution. As each user has secret(private) key, which they do not forward it.
  - Provides key exchange protocols like Diffie-Hellman, ECDHE.
  - Digital Signature can be created using asymmetric encryption, help to authenticate and provide integrity too.
- Disadvantages:
  - Computation Intensive compared to symmetric encryption.
  - Might not suitable for encrypting bulk data as its little slow compare to Symmetric.